Design for performance of broadband signalling and services

D Morris and A M Elvidge

As the telecommunications industry evolves, and broadband services start to arrive, customers are increasingly coming to expect the perception of instantaneous access to service providers, together with transparency to network failures. System performance dictates that response times need to be minimised, sufficient redundant capacity be installed in case of failure, and controls be embedded within the design to manage the exceptional situations (such as media stimulated events) that continually threaten network integrity. A vital part of this system is the broadband signalling network, which underpins the dialogue with the customer, and which enables the delivery of the service.

Network design based on a 'top-down', 'end-to-end' methodology plays a fundamental role in delivering solutions that meet customers' performance needs. Simple, approximate performance models at an early stage in this life cycle are valuable to uncover major performance problems which affect the design of the architecture. Performance issues identified can then be fed back into the design process in an iterative way, to ensure that the design solution will conform to performance requirements.

This paper outlines performance studies of a number of design scenarios for providing broadband services. These studies consider interactive multimedia services, looking at service demand, physical network topology, signalling message flows, the mapping of functional entities to physical components, and routeing, as part of the network design process. The most significant performance issues identified relate to bottle-necks, capacity requirements, and load-dependent response times as perceived by the customer.

1. Introduction

The aim of the performance analysis is to look for areas where the signalling performance of a broadband network, as a whole, may be inadequate. The effect of inadequate performance, as perceived by the user of broadband services, is increased service delays and lower levels of service availability. Long service delays are caused by a greater demand for service than the network can handle. The requests for service are then queued in a buffer which causes the perceived delay.

From the user's perspective the key performance parameter is the service delay. The user is soon frustrated if, after accessing a menu screen to choose their service, the system response time rises to a level where it becomes slow to navigate the menu.

It is most likely that the component with the longest response time is also the most overloaded (unless the network components have been poorly dimensioned), and as such we should be looking, in the first instance, for the 'bottle-neck' in the network. This will point to the component most likely to cause problems by adding the longest delay to the service. When this has been determined the load-dependent delay can be estimated using simple queuing models. In this way a first-cut analysis of the major broadband signalling issues can be carried out, pointing the way to better design and implementation.

2. Assumptions

2.1 Information flows and functional-to-physical mapping

As yet it is unclear what information flows will occur during a broadband multimedia call. There has been some early work to sketch out information flows [1], but these have only dealt with information flows between core network functional entities. The aim was to establish a measure of the quantity of signalling during a broadband call so the performance measures can be estimated at this early stage. The information flows are based on a DAVIC view [2], which is one particular variant on the design of broadband networks.

The information flows are divided into four call processes — initialisation, set-up, transfer and release. Initialisation occurs when the set-top box is powered up. At the end of initialisation, the set-top box has a network address and is able to send signalling messages to the network or a proxy signalling agent (if it requires one). During set-up a user-to-user information stream and an MPEG information stream is established between the set-top box and the level 1 gateway (L1GW). During the transfer phase the set-top box is disconnected from the L1GW and subsequently connected to the interactive multimedia services (IMS) server. A user-to-user information stream and an MPEG information stream is established between the set-top box and the IMS server. Finally a release will occur, which tears down information streams between the set-top box and the IMS server. It should be noted that after the release the set-top box does not need to go through an initialisation stage in order to set up a new broadband connection. This only has to be done after the set-top box has been powered down. The information flows in Figs 1 — 4 show the functional entities of the network. (Note that information flows containing a proxy signalling agent are not shown.)

The information flow marked as ‘user data’ is the point in a call where the set-top box and the level 1 gateway or IMS server exchange DSM-CC user-to-user signalling messages and an MPEG video stream.

The names given to the information flows shown in Figs 1 — 4 are followed by a number in brackets. This number represents a best estimate of the actual number of messages involved in supporting the information flow and it is included because, from a performance point of view, it is the number of messages which is of importance. The estimate of the actual number of messages used is based on the format of the Q.2931 protocol.

2.2 *Network architectures*

In the early stages of broadband network design, there is no one definitive design on which to base a broadband network architecture. Thus, from a performance point of view, rather than there being a single network scenario to assess, a number of variants need to be considered. In this study the number of network architectures has been reduced to two — the session manager and L1 gateway in a single network node, and the session manager, L1 gateway and proxy signalling agent in separate network nodes. The aim of these configurations is to test how much processing is required at a physical network node. Putting more functionality into a single node will generally decrease its performance, since each supported function requires a finite amount of processing. On the other hand, separating functionality increases demand for installed signalling and user bandwidth.

General assumptions made about the scenarios in the context of interactive multimedia services are:

- the session manager (SM) is located within the ATM switch,
- the ATM switch signals to some network intelligence platform (NIP) using signalling similar to the intelligent network application protocol (INAP),
- the L1GW is an IP which provides video dial tone and collects user selection information,
- the L1GW is not involved in the call beyond the call set-up phase,
- the NIP directs the call set-up between the L1GW and the set-top box, or the IMS server and the set-top box, using signalling similar to INAP,
- the proxy signalling agent (PSA) is not located in the access network (i.e. at the VADSL access point (VAP)).

Figures 5 and 6 show the two network architectures considered.

2.3 *Number of visits*

The message sequence charts provide an indication of how busy various network components will be for a given network architecture and call procedure. This is done by counting the number of times each network component is used, or visited, for each call procedure. To maintain a consistent view a visit occurs when a signal is sent **or** received at a network node. The reasoning here is that processing is required when a signalling message is either sent or received. Tables 1 and 2 summarise the number of visits to each network component.

However, knowing the number of visits to each component is not enough as we need to know that functional-to-physical mapping exists. Put another way, each signal sent or received has to be processed and where that processing is carried out can have an effect on system performance. Table 3 shows the number of visits each network node experiences given the two scenarios assumed.

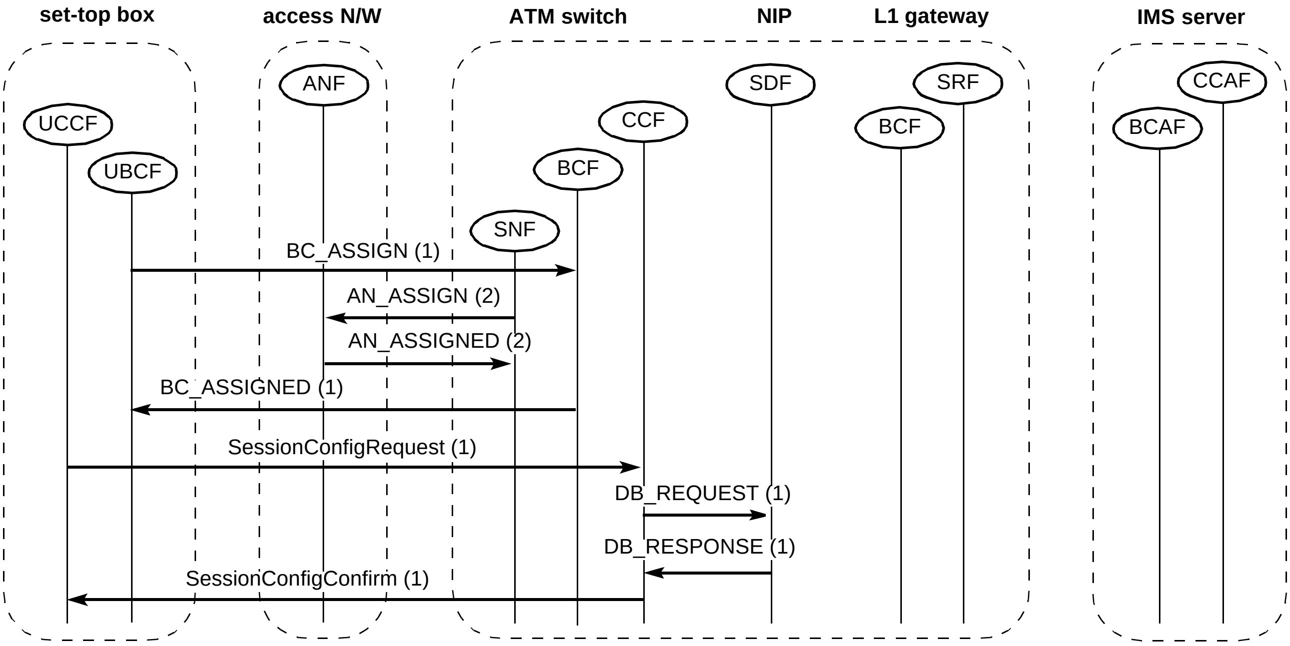


Fig 1 Session initiation.

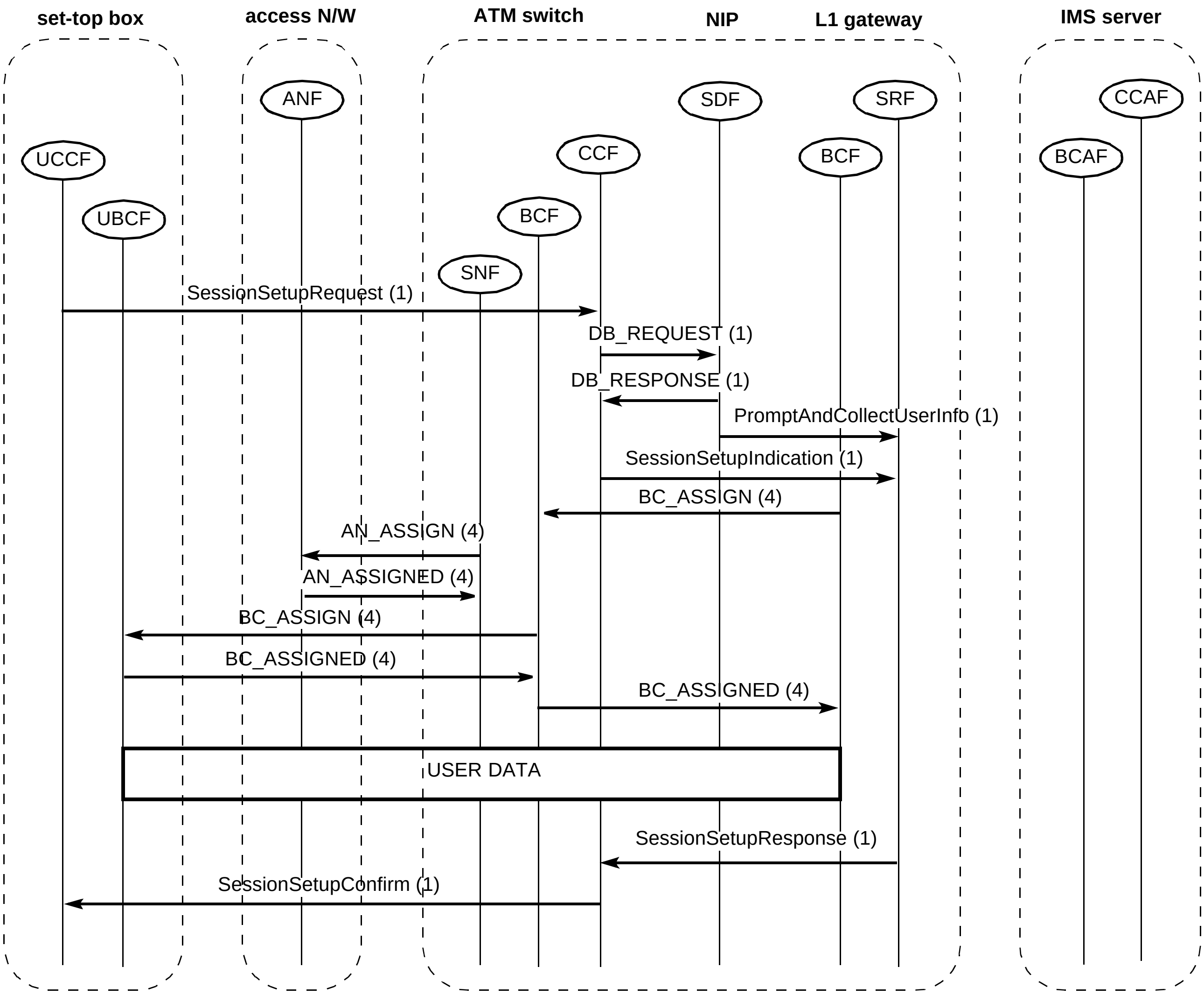


Fig 2 Session set-up.

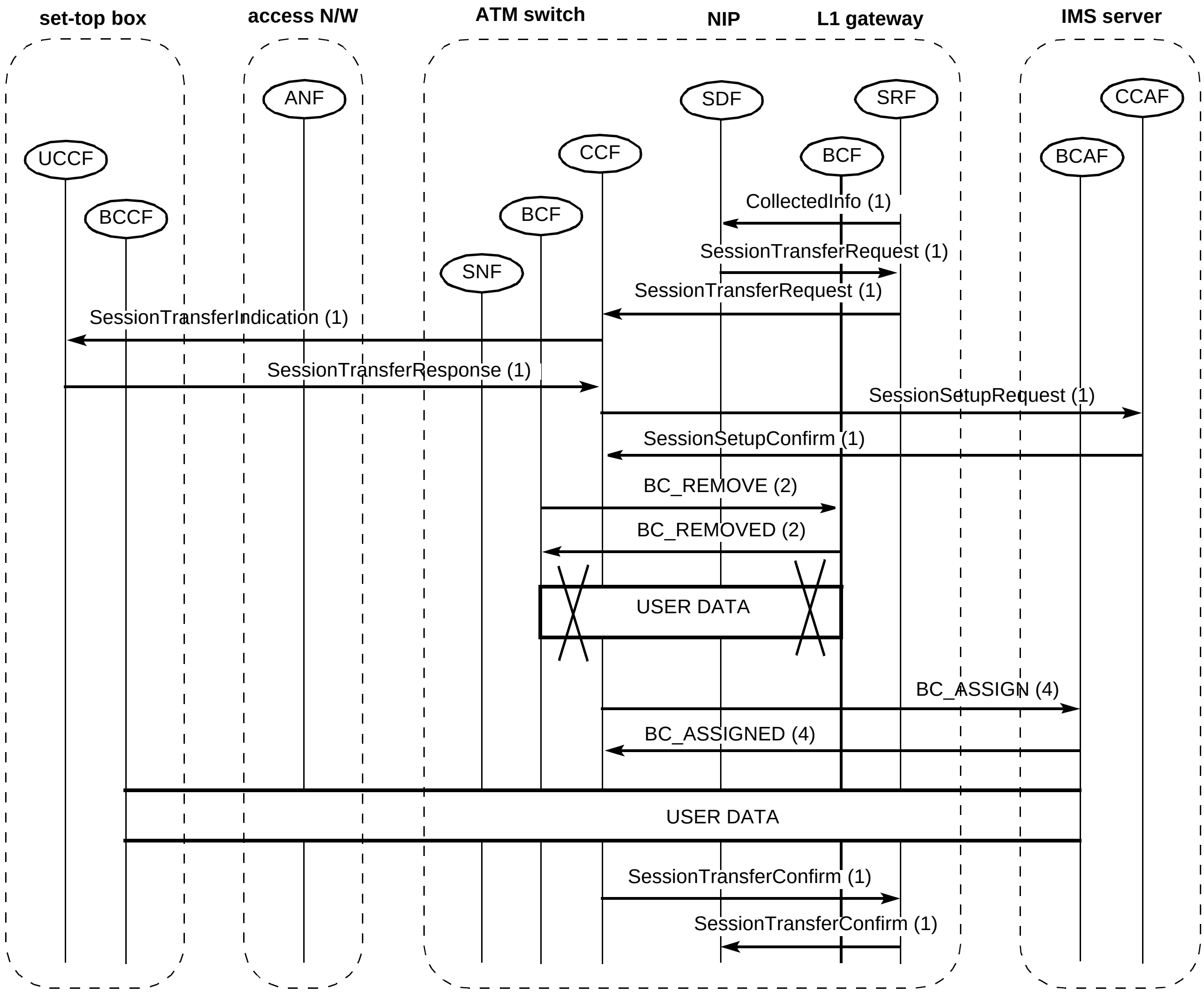


Fig 3 Session transfer.

Table 1 Number of visits with no proxy signalling agent.

	STB	SM	LIGW(IP)	NIP	IMS	AN
Initiation	4	10	0	2	0	4
Set-up	10	30	11	3	0	8
Transfer	2	18	9	3	10	0
Release	6	17	0	1	6	4
Total	22	75	20	9	16	16

Table 2 Number of visits with proxy signalling agent.

	STB	SM	LIGW(IP)	NIP	IMS	PSA	AN
Initiation	6	16	0	2	0	4	4
Set-up	3	36	11	3	0	13	8
Transfer	2	18	9	3	10	0	0
Release	3	21	0	1	6	7	4
Total	14	91	20	9	16	24	16

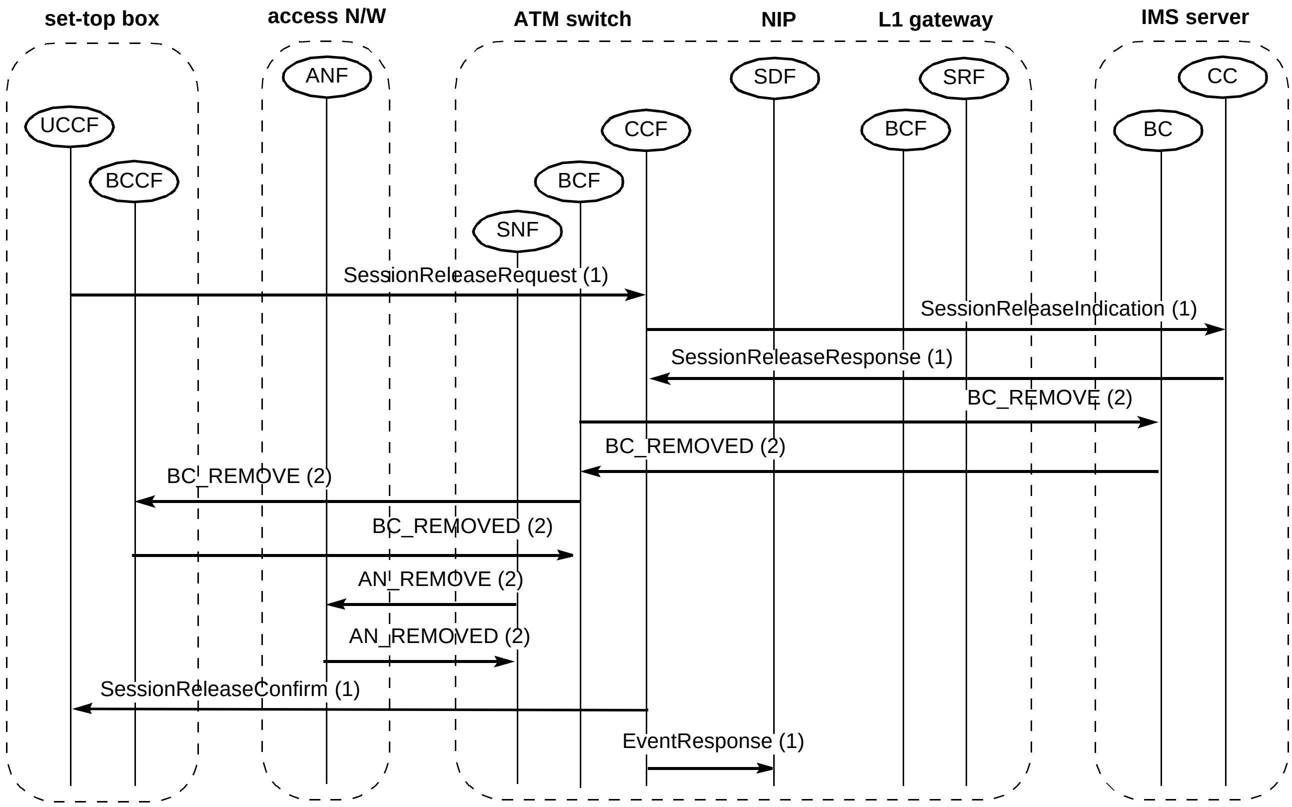


Fig 4 Session release.

Table 3 Number of visits to each physical node (I — initialisation, S — set-up, T — transfer, R — release).

	{L1GW, SM}					{L1GW}, {PSA}, {SM}				
	I	S	T	R	Total	I	S	T	R	Total
Switch node	10	41	27	17	95	16	36	18	21	91
L1GW node					0	0	11	9	0	20
STB node	4	10	2	6	22	6	3	2	3	14
NIP node	2	3	3	1	9	2	3	3	1	9
IMS node	0	0	10	6	16	0	0	10	6	16
PSA node					0	4	13	0	7	24
AN node	4	8	0	4	16	4	8	0	4	16

2.4 The bottle-neck

The bottle-neck is the network component which is most highly utilised. This component will handle the lowest number of calls per second and will be the first to overload. To ascertain which component is the bottle-neck, the number of visits per call made to that component, the average service time for each server, and the number of servers that component has, must all be known. (Note that for a queuing system the service time is the time it takes to service a message once it reaches the head of the queue.) The maximum capacity of that network component is represented as:

$$\text{Maximum call-handling capacity} = \frac{N}{S \times V} \text{ calls/s,}$$

where:

N is the number of servers,

S is the average service time for each server,

V is the total number of visits (to all servers).

The bottle-neck is then by definition the component with the lowest maximum capacity.

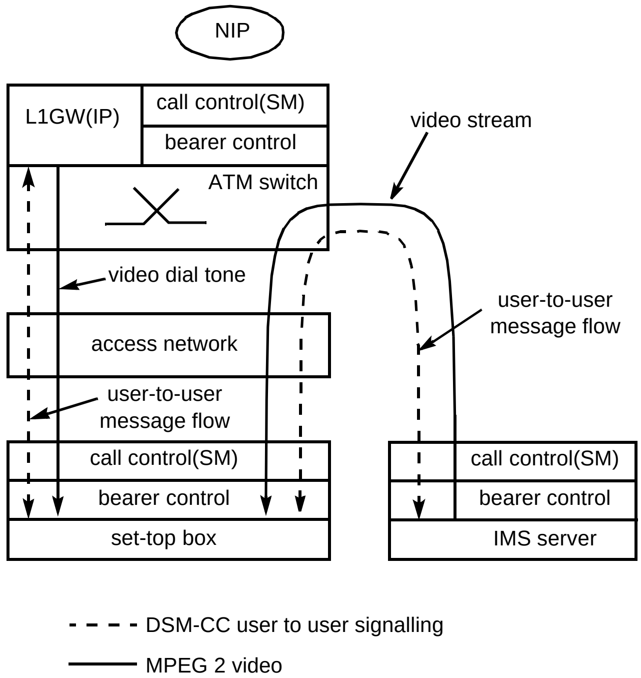


Fig 5 L1GW and SM in a single network node.

However, this is a theoretical capacity unachievable in real systems. The maximum call-handling capacity is the capacity at which the response time rises to infinity, where the response time is the total of the queuing time and the service time. To achieve reasonable response times and to allow for fluctuations in traffic network, nodes are typically operated at a fraction of the maximum call-handling capacity.

In fact, there is not actually enough information at the present moment in time to determine the bottle-neck because the average service time per visit for most of the network components is not known — these components have yet to be designed and built. From a performance point of view, this is a problem because, without knowing where the bottle-neck is, it is not possible to analyse the system as a whole. So an assumption must be made, which is that the ATM switch node will be the bottle-neck. This seems reasonable given that the session manager which is assumed to reside on the switch experiences the most number of visits (see Tables 1 and 2) and that the complexity of the operations processed by the session manager are high since it is handling call and bearer control. However, it has to be borne in mind that this is an assumption and in fact any network component could be the bottle-neck.

The average service time per visit depends on the make of the ATM switch and its processing characteristics. For the purposes of this study, it is assumed that the average service time per visit is 4 ms [3].

2.5 *Load-dependent response time*

All components in a network will have a load-dependent response time, but only the bottle-neck will experience a dramatic rise in response time as the system reaches its maximum capacity. Response time assumptions are shown in Table 4.

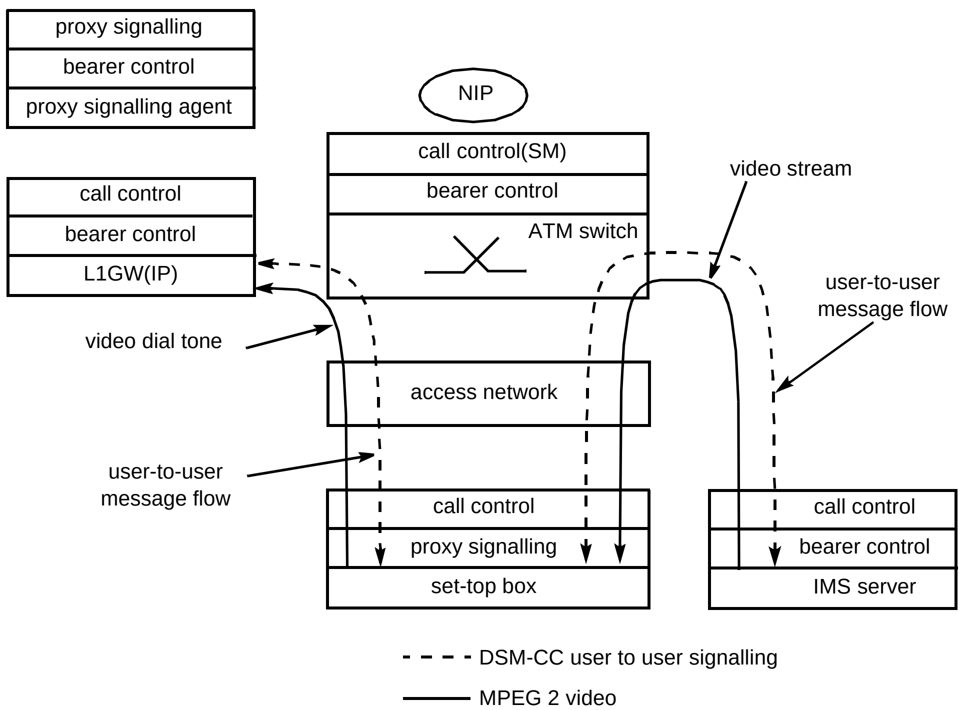


Fig 6 L1GW, PSA and SM all in separate network nodes.

Table 4 Response time per visit.

Independent network component	Response time per visit (ms)
STB	10
SW	Load dependent
PSA	10
L1GW	10
NIP	25
IMS	10
AN	10

The queuing model for the ATM switch was assumed to be M/D/4 with an additional fixed delay, following an analysis of narrowband switch behaviour for narrowband calls [3].

3. Performance issues

3.1 System capacity

The term ‘maximum system capacity’ is defined in this paper as the maximum rate at which the system can handle broadband calls. The bottle-neck determines maximum system capacity and so it is the capacity of the ATM switch node that needs to be resolved. If it is assumed that a complete call requires set-up, transfer and release call processing, the load on the ATM switch node can be estimated by noting the figures contained in Table 3 and remembering that the average service time per visit for the ATM switch is assumed to be 4 ms. In order to compute system capacity the ratio of the number of broadband calls to the number of set-top box initialisations made needs to be specified. Here it will be assumed that for every three broadband calls, one set-top box initialisation occurs.

Maximum system capacity (as defined mathematically) is the capacity at which the response time rises to infinity. This is not an achievable operating point. In this paper a capacity figure which is a fraction of maximum system capacity is termed the ‘normal capacity’ figure, and it is this figure which is assumed to be the operational capacity of the system (see Table 5 [3]).

Table 5 Capacity of switch node.

Scenario	No of visits to switch node per call	Normal capacity (calls/second)
{L1GW, SM}	88.3	1.36
{L1GW}, {PSA}, {SM}	80.28	1.49

3.2 Demand on NIP

In this paper, broadband calls are intelligent network (IN) calls, and as such require signalling co-ordination functionality [4]. The assumption is that this will be provided by a network intelligence platform.

Table 5 provides the maximum capacity of the switch node for each functional-to-physical mapping, and from Table 3 it can be deduced that for any call the NIP is visited nine times. This corresponds to an average of 4.5 NIP transactions per call, noting that a transaction normally requires two visits (one message received and one message sent). Thus:

calling rate at the NIP = normal calling rate of each switch × number of switches

transaction rate at the NIP = 4.5 × calling rate at the NIP

The NIP transaction rate is shown in Table 6.

Table 6 The normal NIP transaction rates.

Scenario	Normal calling rate at the NIP (calls/ second)	Normal NIP transaction rate (transactions/ second)
{L1GW, SM}	67.95	306
{L1GW}, {PSA}, {SM}	74.74	336

3.3 Response time

For queuing systems with a finite service time, the response time, i.e. the time for a system to process a transaction, generally increases with increasing load.

If the response time for component i is R_i then the total response time for a system is generally expressed as:

$$\text{total response time} = \sum_i R_i V_i$$

where V_i is the number of times the ith component is visited. Given that the number of visits to each physical node is known from Table 3, and the assumed response times for each non-bottle-neck node is known from Table 4, the load-independent response time for each call procedure can be calculated. Assuming the M/D/4 model with additional fixed delay for the bottle-neck node, i.e. the switch node, and noting that the service time for each of the 4 servers in this node is 16 ms per visit (i.e. an average service time of 4 ms per visit across all servers), an overall load-dependent response time can be computed for each call procedure in each scenario. These are shown in Figs 7 and 8.

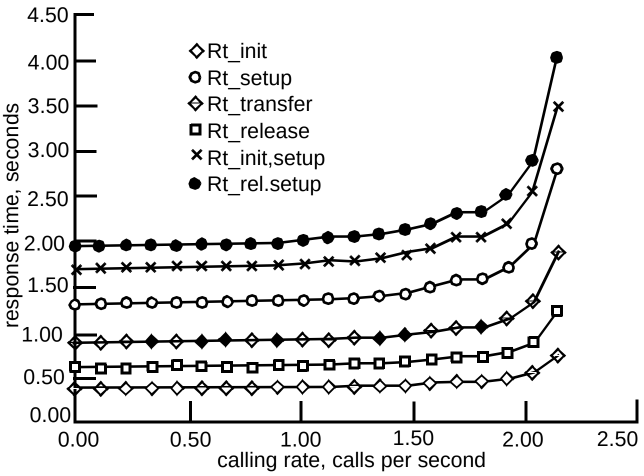


Fig 7 Response time for {L1GW, SM}.

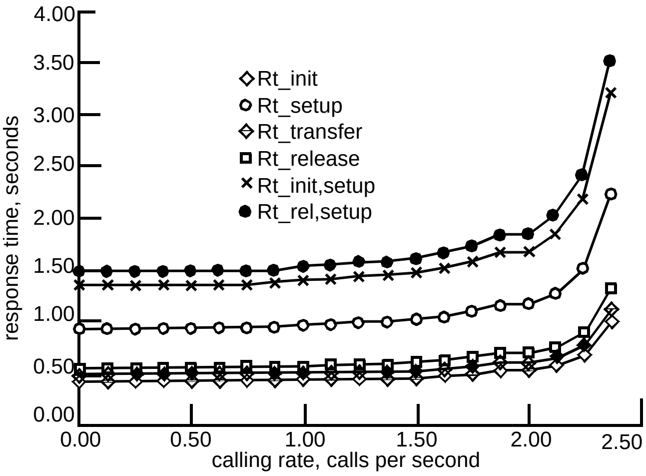


Fig 8 Response time for {L1GW}, {PSA}, {SM}.

The legends in Figs 7 and 8 are as follows:

Rt_init — response time for initialisation call procedure,

Rt_setup — response time for set-up call procedure,

Rt_transfer — response time for transfer call procedure,

Rt_release — response time for release call procedure,

Rt_init,setup — total response time for initialisation and set-up call procedures,

Rt_rel,setup — total response time for release and set-up call procedures.

In all cases the single call procedures (initialisation, set-up, transfer and release) completed within 1.5 sec at normal load. The double call procedures (initialisation/set-up and release/set-up) completed in under 2 sec at normal load.

4. Discussion

The aim of this paper is not necessarily to provide exact answers on topics like response time and system capacity, but rather to point out some major issues which will need careful consideration as the specification of the broadband access network accrues more detail. At present the specification of a broadband access network has yet to be completed. This gives the performance engineer the freedom to carry out a comparative study of a number of different scenarios in order to test which is likely to be the best.

The capacity of each broadband switch was found to be in the range 0.99 to 1.76 calls per second with the assumptions made. This can be compared with a single cluster System X unit which can handle up to about 25 ordinary telephony calls per second. This difference is a consequence of a broadband call requiring more messages to complete. A complete telephony call requires about 7 messages, and a narrowband call using network intelligence, such as a OneNumber call, requires up to 16 messages, whereas a broadband call may require anywhere between 138 and 158 messages (excluding initialisation) depending on the network architecture. What should be noted is that even at this early stage of specification it is clear that broadband networks will require significantly more signalling and thus processing power. This should not be overlooked when considering the performance of the final system design, or when designing the message flows to support the service.

In this paper the design of the broadband network is assumed to follow IN principles. User and IMS server profiles and set-top box information is assumed to reside in a NIP. This means that the NIP is involved in a broadband call. Also, the level 1 gateway is assumed to be an IP which requires direction from the service control function which also resides in the NIP. First considerations suggest that the NIP is required to carry out about 4.5 transactions per broadband call. Given a knowledge of the predicted customer base and possible customer calling rates, the total demand on the NIP for transactions per second can be calculated, and this information then used to dimension a NIP.

An attempt has been made to assess the load-dependent response time for a broadband network. Two important points come out of this. Firstly, what affects the user's perception of delay is not necessarily the total response time for all call procedures' initialisation/set-up, transfer and release, but the delay at each of the call procedures, in particular the initialisation/set-up delay and transfer delay. The second point is that delay rises dramatically if system capacity is approached, and so, for acceptable service to be offered at normal loads, the system dimensioning needs to take this into account.

5. Conclusions

A broadband call may require anywhere between 138 and 158 messages (excluding initialisation) compared to 7 messages for a standard telephony call and around 16 messages for a narrowband OneNumber call. The signalling traffic generated is therefore set to increase dramatically, with implications for both customer perceived delays and signalling network dimensioning and financial investment. Signalling flows need to be minimised to offer acceptable quality at a reasonable price.

The technique discussed in this paper allows the switching capacity per broadband switch to be identified, in calls per second at normal load, i.e. when the system is running at considerably less than maximum capacity.

First considerations suggest that the NIP is required to carry out about 4.5 transactions per broadband call, and so, given the predicted customer base and possible maximum customer calling rates, the demand offered by the broadband network to a NIP, and hence the required NIP capacity, can be established. The call set-up and transfer delay perceived by the customer can also be calculated, given the network architecture and normal demand for the broadband service. This information is vital in dimensioning equipment levels to yield system delays which are acceptable to the customer.

Finally, response times rise dramatically if system capacity is approached, and therefore the system needs to be dimensioned so that the occupancy of the bottle-necks is considerably less than 100% at normal load.

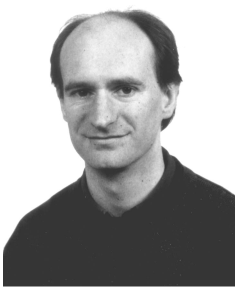
References

1 Jones I G and Reece P W: 'Proposed information flows', 3732/T/0126, Draft A (December 1995).

2 ISO: 'Digital storage media command and control', IEC 13818-6, Boston (May 1995).

3 Morris D and Elvidge A M: 'Initial performance analysis for ADP1', DSN2:3732:NIPT Memo 407/95 (3732/141), Issue 1 (March 1996).

4 Knight R R: 'Signalling and IN for T&DC', 95/3732/074, Issue 1 (November 1995).



Dave Morris received an MSc in Opto-electronics and Information Theory from Queen's University of Belfast in 1988. Prior to this he obtained a BSc in Physical Electronics in 1986. Since 1988 he has been working at BT Laboratories, initially on the design of high-speed optical modulators and then on 3rd generation mobile systems.

He currently works on the performance analysis and dimensioning of narrowband and broadband networks.

Ann Elvidge joined BT Laboratories following her PhD at the University of Surrey in 1985, and initially worked on voice network design and performance. Following a secondment to Syncordia, she moved into modelling SS No 7 and broadband signalling networks. She currently represents BT at the ITU-T on network operations, and following completion of her MBA at Henley Management College in 1996, she is active in business modelling, providing support for business cases for new services.